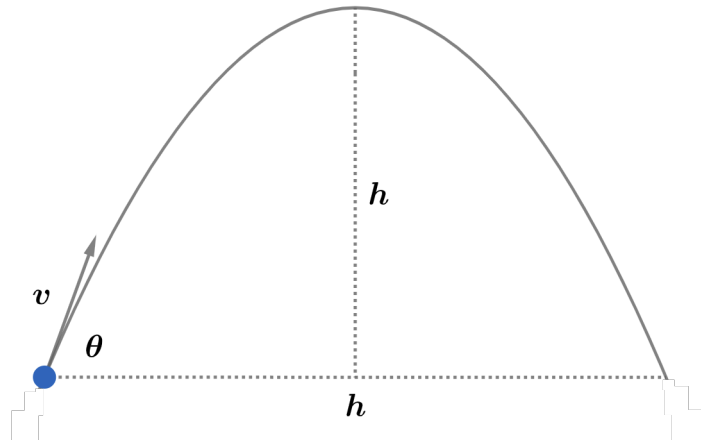


2023 F=ma Exam: Problem 11

Kevin S. Huang



Recall the range and height equations,

$$R = \frac{v^2 \sin(2\theta)}{g}$$

$$H = \frac{v^2 \sin^2 \theta}{2g}$$

Since $R = H$,

$$\frac{v^2 \sin(2\theta)}{g} = \frac{v^2 \sin^2 \theta}{2g}$$

$$2 \sin \theta \cos \theta = \frac{\sin^2 \theta}{2}$$

$$\tan \theta = 4$$

Because we reach height h ,

$$h = \frac{v^2 \sin^2 \theta}{2g} = \frac{v^2 \sin^2(\arctan 4)}{2g} = \frac{8v^2}{17g}$$

$$v^2 = \frac{17gh}{8}$$

If we launch with this speed upwards, then by energy conservation,

$$\frac{1}{2}mv^2 = mgh'$$

$$h' = \frac{v^2}{2g} = \frac{17h}{16}$$

so the answer is A.